

9(a)	<i>any six from:</i> 1 add marker gene to the, vector / plasmid ; 2 gene of interest inserted close to marker gene ; 3 (marker) gene product / protein, emits light ; 4 visible colour change ; 5 <i>ref. to</i> exposing to UV light / laser scanner ; 6 easy to identify transformed, bacteria / organisms ; 7 examples ; e.g. GFP 8 <i>idea of</i> no known risk ; 9 AVP ; e.g. <i>ref. to</i> gene of interest inserted into marker gene / insertional inactivation	6
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Question	Answer	Marks
9(b)	<i>any five from:</i> <i>advantages</i> 1 increase in yield ; 2 improved quality / AW ; 3 improvement to health / Golden rice™ and vitamin A deficiency ; 4 longer shelf-life ; 5 some GM crops are adapted to unfavourable conditions ; A e.g. drought tolerance / nitrogen fixing / salt tolerance 6 (insect / herbicide, resistant crops) so less money spent on, pesticide / herbicide ; <i>any five from:</i> <i>disadvantages</i> 7 consumer resistance to GM crops ; 8 may be unsafe for humans / allergies / side effects / harm other animals ; 9 expensive ; 10 may have to buy seeds every season ; 11 seed and related herbicides sales monopolised by big companies ; 12 <i>ref. to</i> affects organic food ;	9